

Lab Assignment 2: Regression: Housing Price Prediction

CSAI2017P: Applied Machine Learning (Lab)

Experiment 3 (Unit V) • CO2, CO4

Instructor: Dr. Tofik Ali

Due: announced in the lab

Student Name: _____

SAP ID: _____

Batch: _____

Lab quiz: LQ2, after submission

Objective

- Fit simple and multiple linear regression and read the coefficients.
- Evaluate a regressor with metrics that carry the units of the target.
- Recognise over-fitting and control it with regularization.
- **Before the lab:** run `fetch_california_housing()` once at home. It downloads ~1.4 MB and the campus network may block it.

Datasets used in this assignment

A1 (California Housing).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA02_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)

A1. Simple linear regression

[2 marks]

Dataset: A1. Target: MedHouseVal.

- (a) Fit a regression of MedHouseVal on MedInc alone.
- (b) Report the slope, the intercept and test R^2 .
- (c) State in one sentence what the slope means in real units.

A2. Multiple linear regression

[2 marks]

Dataset: A1, all features.

- (a) Fit a multiple linear regression on all eight features inside a scaling pipeline.
- (b) Report test MAE, RMSE and R^2 .
- (c) Plot predicted against actual and describe the pattern in two lines.

A3. Metric choice

[2 marks]

Dataset: A1.

- (a) Compute MAE, MSE and RMSE for your model from A2.
- (b) State which of the three is hardest to explain to a non-technical client, and why.
- (c) State which you would report if a few very expensive districts matter most, and why.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)

B1. Polynomial features and over-fitting

[4 marks]

Dataset: A1.

- (a) Build pipelines with PolynomialFeatures of degree 1, 2 and 3 followed by Linear-Regression.
- (b) Report train and test RMSE for each degree in one table.
- (c) Identify the degree at which over-fitting begins and explain in 4–6 lines how the table shows it.

B2. Ridge and Lasso

[4 marks]

Dataset: A1 with degree-2 polynomial features.

- (a) Fit Ridge and Lasso for $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ and plot test RMSE against α .

- (b) Report the best α for each and the count of exactly-zero coefficients in the best Lasso.
- (c) Explain in 4–6 lines why Lasso zeroes coefficients and Ridge does not.

Section C (Hard) — compulsory (2)

C1. Where the model fails

[2 marks]

Dataset: A1.

- (a) Take the 20 test districts with the largest absolute error from your best model.
- (b) Describe in 4–6 lines what they have in common and what you would add to the data to fix it.

End of Lab Assignment 2

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.